Engineering Mathematics - I · Unit_02 · Unit 2

DRAFT — UE26MA141A · Unit_02 · Unit 2 · Particular integral Type 1 and Type 2

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Particular integral Type 1 and Type 2

1 Particular integral Type 1 and Type 2

Study focus: Particular integral_Type 1 and Type 2. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000066 · EV0000067

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we solve problem on finding the particular integral when capital X is k power X and X equals.

Key details

- Sine of AX plus B or cos of AX plus B.
- You have already learned how to find Pi when right hand side capital X is e power AX, where a can be positive, negative, or even zero.
- over f of a it means we will replace d by a in f of d provided f of a has to be not zero in case f of a becomes zero as from theory the particular integral is x.
- Expressed in this particular form that is x times e power ax divided by f dash of a.
- It means we differentiate f of d with respect to d once and again replace d by a again f dash of a should be not zero.
- In case f dash f a becomes zero, so the particular integrity will be modified in this particular form.
- That is x square times e power a x divided by f double dash a and so on.
- Observe carefully the right hand side is one.
- This is nothing but e power 0x where a is taken as 0.
- Replace d by 0, we get 0 plus 0 plus 5, so 1 by 5.
- So because this is non-zero, we can stop here.
- P particular integral PI write the formula one over.
- D square plus four d plus four times a power minus two x.
- Observe carefully, we need to replace d by a when a is minus two.
- Again we see that denominator is zero when d is minus two.

- Therefore this is written as x square times one over two d becomes two four becomes zero e power minus two x.
- The right hand side is sine hyperbolic X.

Example or application from the lecture

- Let us solve problems.
- Take an example of finding the particular integral for this different equation.
- Let us take another example where the right hand side is a power minus 2x.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

Core explanation · 05:00–09:59

Therefore this is same as e power x minus e power minus x.

Key details

- As you can see this is P1 and P2.
- Even in the second stage, f dash f is also zero.
- Therefore, P becomes x square times e power x divided by 6.
- Whereas in the.
- Second case denominator is non-zero.
- Finally Pi is P1 minus P2.
- When x is sine of a x plus b or cos of a plus b.
- Remember, if we want to apply this case, there are certain constraints.
- F of D should have D square in it.
- Constraint number two.
- We replace D square by minus A square.
- Remember this is not minus A whole square, it's minus A square.
- to get d square in f of d.
- That is why when you write the particular integral this is not f of t this is f of d square.
- And we are replacing d square by minus a square, provided this has to be non-zero.
- In case it becomes zero, we go by case one.
- Introduce one x differentiate f of d bet.
- Respect to D once with respect to D1s.
- Again, that derivative should have d square, then replace d square by minus a square, and so on.
- Again this is minus a square.
- Find the particular integral for this differential equation where the right hand side is cos of row 3x.
- And you can see a is root three.
- Minus three whole square.
- Therefore this is minus three.
- Therefore denominator becomes zero.
- In this case Pi is expressed in the form x into derivative of d squared becomes two d.

- Take x by two side integrate cos root 3x with respect to x, then we arrive at the result.
- X by 2 sine root 3 x divided by root 3.
- So this problem should be solved by both case 1 and case 2 because for this one we use e ball 0x and for this we use case number 2.

Example or application from the lecture

- Let us apply case one two times.
- Let us move on to second case.
- Let's say let us take an example.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 10:00–14:59

It's very easy to see that when you compute d1, when you replace d by gate, denominator becomes zero.

Key details

- Then we introduce one x.
- I differentiate with respect to D this 3D square.
- This is minus 40 and this is 1.
- Again 3d square becomes 0 and 4d becomes 0, the left out number is 1.
- Therefore, the answer for this P1 is simply x by 2.
- How do we solve P2?
- And you observe carefully as per theory part this f of d doesn't have d square.
- It means multiplying divided by the conjugate of 8 minus 3d.
- It means 8 plus 3d of cost 2x divided by 8 minus 3d into 8 plus 3d.
- That's going to be 64.
- Minus 36 B square.
- In the denominator,.
- When you replace d square by minus 4, you get 100.
- Sine of Ax plus B or cos of X plus B.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Particular integral_Type 1 and Type 2 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_Type_1_and_Type_2__54bec5745a32a447.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral_Type 1 and Type 2 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_Type_1_and_Type_2__54bec5745a32a447.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral_Type 1 and Type 2 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_Type_1_and_Type_2__54bec5745a32a447.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9979; only the JSON segmentation supplies the displayed timestamps.
Particular integral_Type 1 and Type 2 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_Type_1_and_Type_2__54bec5745a32a447.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.

5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00014	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0046. Sidecar variants compare...
SRC00015	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0046. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00016	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0046. Sidecar variants compare...
SRC00017	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0046. Sidecar variants compare...

C Revision and retrieval practice